



Preparing for Sequencing

Designing Life with AI

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Description

Sequencing is performed to determine the exact nucleotide sequence of the amplified DNA fragment and confirm if the targeted sequence is present and correct.

Steps

I – Clean PCR products

This first step is performed using the Monarch PCR & DNA Clean-up Kit following the manufacturer's instructions:

<https://www.neb.com/en/products/t1030-monarch-pcr-dna-cleanup-kit-5-ug>

Protocol: All centrifugation steps should be carried out at 13'000 rpm.

1. Dilute sample with DNA Cleanup Binding Buffer according to the table below. Mix well by pipetting up and down or flicking the tube. Do not vortex.
2. Insert column into collection tube and load sample onto column. Spin for 1 minute, then discard flow-through.
3. Add 200 µl DNA Wash Buffer and spin for 1 minute.
4. Repeat wash (Step 3).
5. Transfer column to a clean 1.5 ml microfuge tube.
6. Add 15 µl of ddH₂O. Wait for 1 minute, then spin for 1 minute to elute DNA.
7. Perform a DNA concentration and purity measurement with a nanodrop machine. If DNA purity is too low, perform the washing steps again.

Sample type	Ratio of binding buffer : sample	Example
dsDNA > 2 kb (plasmids, gDNA)	2:1	200 µl : 100 µl
dsDNA < 2 kb (some amplicons, fragments)	5:1	500 µl : 100 µl
ssDNA > 200 nt	7:1	700 µl : 100 µl

II – Sample preparation

1. In a 1.5 ml Eppendorf tube, mix 12 µl of cleaned DNA with 3 µl of primer (one primer per tube).
2. Elute the mixture in 35 µl of distilled water.
3. Place a Microsynth sticker on each tube to label and number your sample.
4. Fill up all the information and send for Sanger sequencing.